A DDR Scheme for the Elliptic Problem with Interface

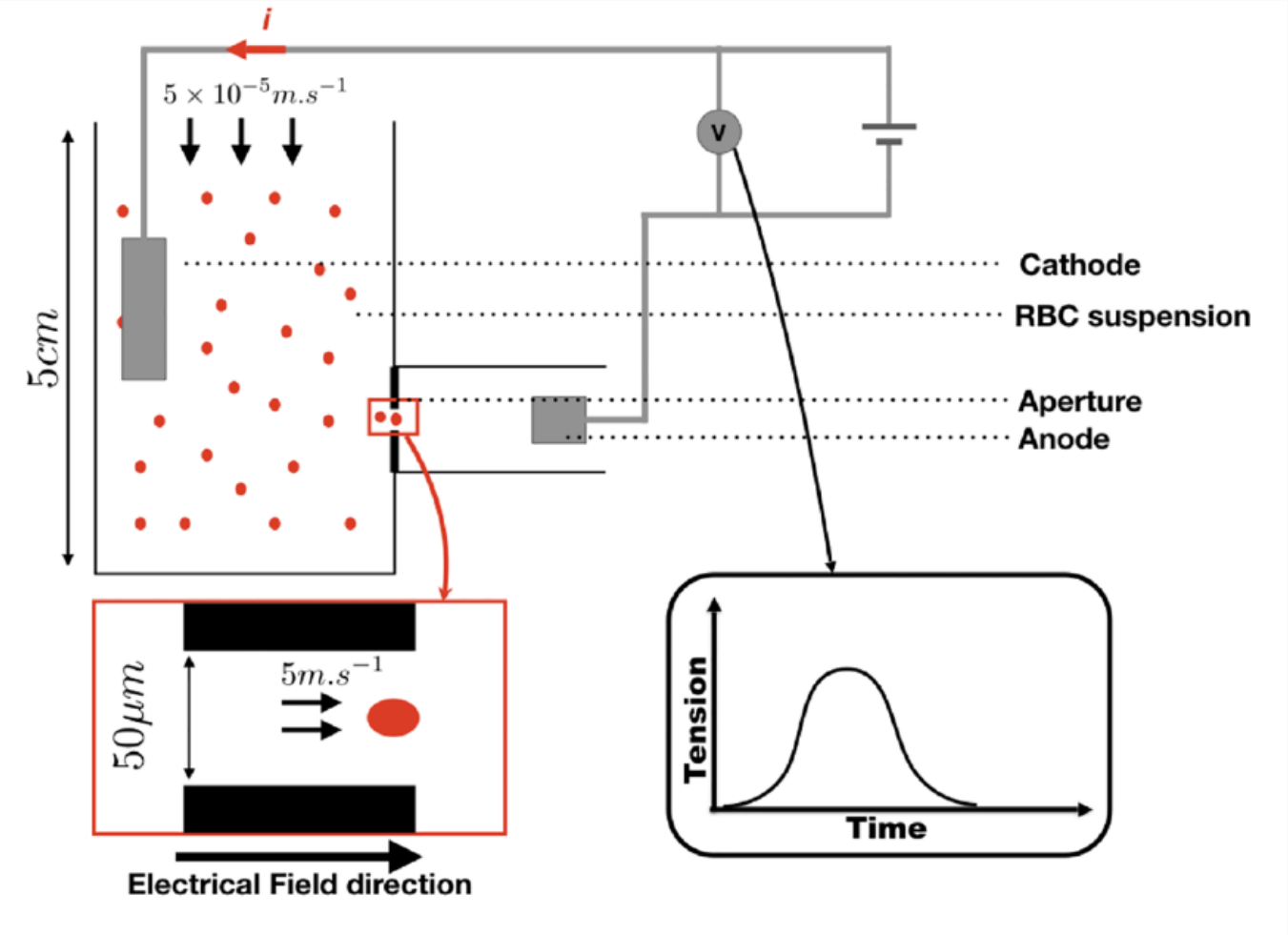
Estimation of the Interfacial Electric Field for a biological membrane

February 7, 2025

Outline

- Engineering Motivation
- Stationary Interface Condition
- Unsteady Interface Conditions
- Moving Interface

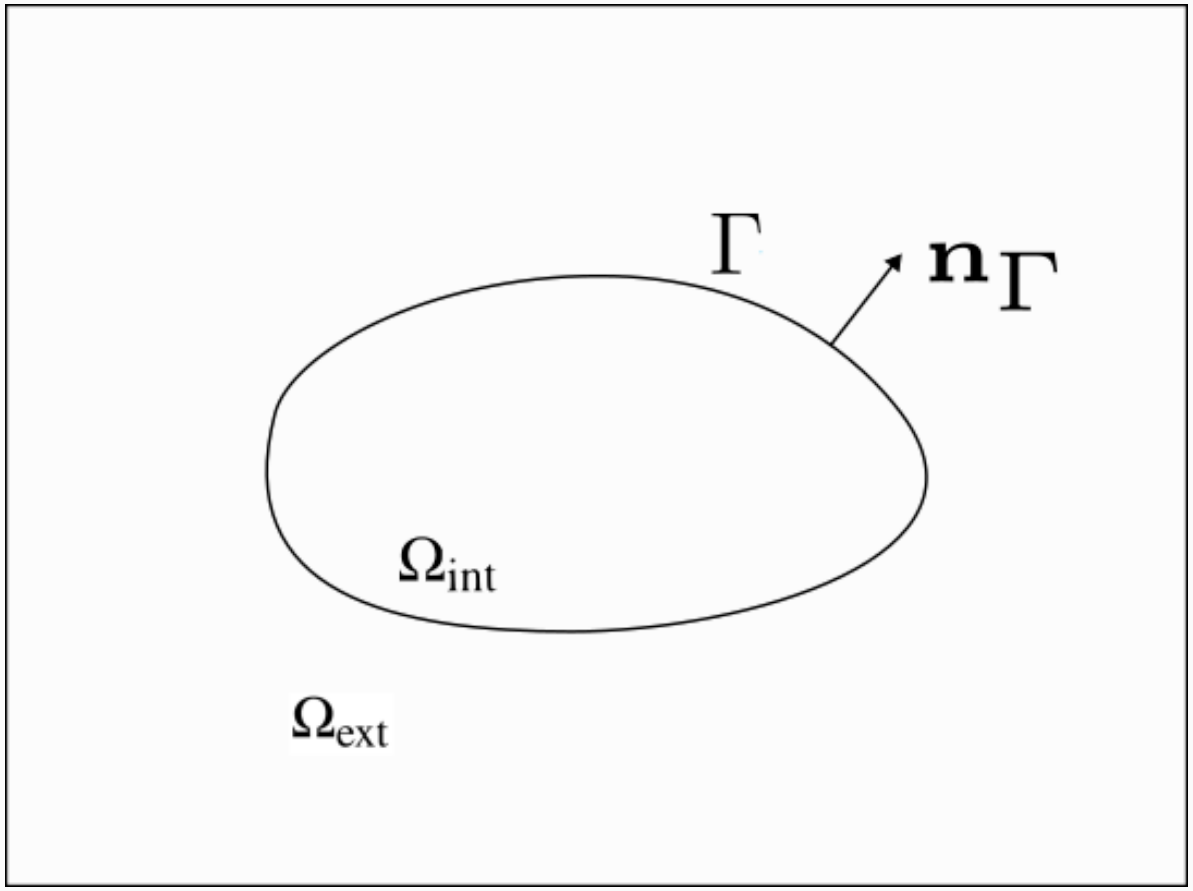
Coulter Counter



Domain and Interface

Domain $\Omega \in \mathbb{R}^d$,

Interface Γ: closed $d-1$ surface, with outward normal n_Γ.



In the following: $d = 2$.

Model

Given: $\sigma : \Omega \setminus \Gamma \to \mathbb{R}$ s.t. $\sigma_{|\Omega_{\mathrm{int}}} \equiv \sigma_{\mathrm{int}} > 0$ and $\sigma_{|\Omega_{\mathrm{ext}}} \equiv \sigma_{\mathrm{ext}} > 0$;
$f : \Omega \setminus \Gamma \to \mathbb{R}$, $\Phi_\Gamma : \Gamma \to \mathbb{R}$, and $J_\Gamma : \Gamma \to \mathbb{R}$;
find $u = (u_{\mathrm{int}}, u_{\mathrm{ext}})$ with $u_\bullet : \Omega_\bullet \to \mathbb{R}$, $\bullet \in \{\mathrm{int}, \mathrm{ext}\}$ such that:

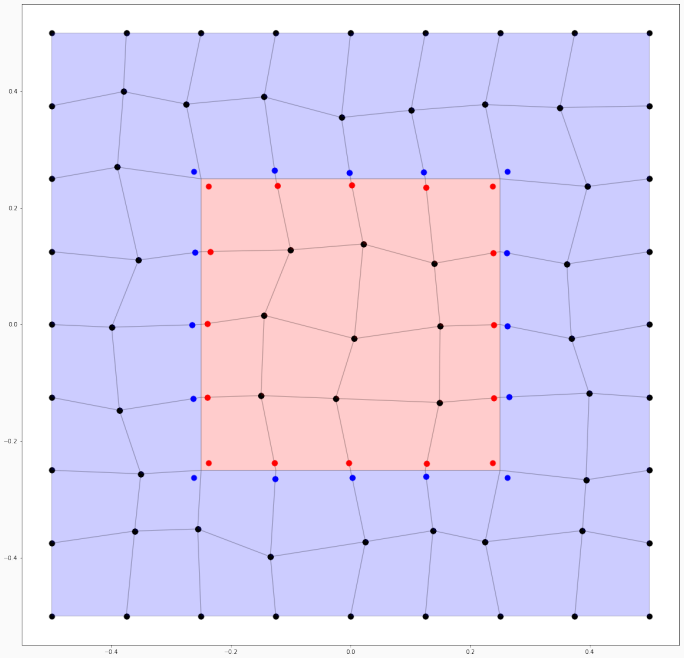
$$
\begin{aligned}
-\nabla \cdot (\sigma_\bullet \nabla u_\bullet) &= f && \text{in } \Omega_\bullet,\ \bullet \in \{\mathrm{int}, \mathrm{ext}\}, && \text{(1a)}\\
[u]_\Gamma &= J_\Gamma && \text{on } \Gamma, && \text{(1b)}\\
[\sigma \nabla u]_\Gamma \cdot n_\Gamma &= \Phi_\Gamma && \text{on } \Gamma, && \text{(1c)}\\
u_{\mathrm{ext}} &= 0 && \text{on } \partial\Omega, && \text{(1d)}
\end{aligned}
$$

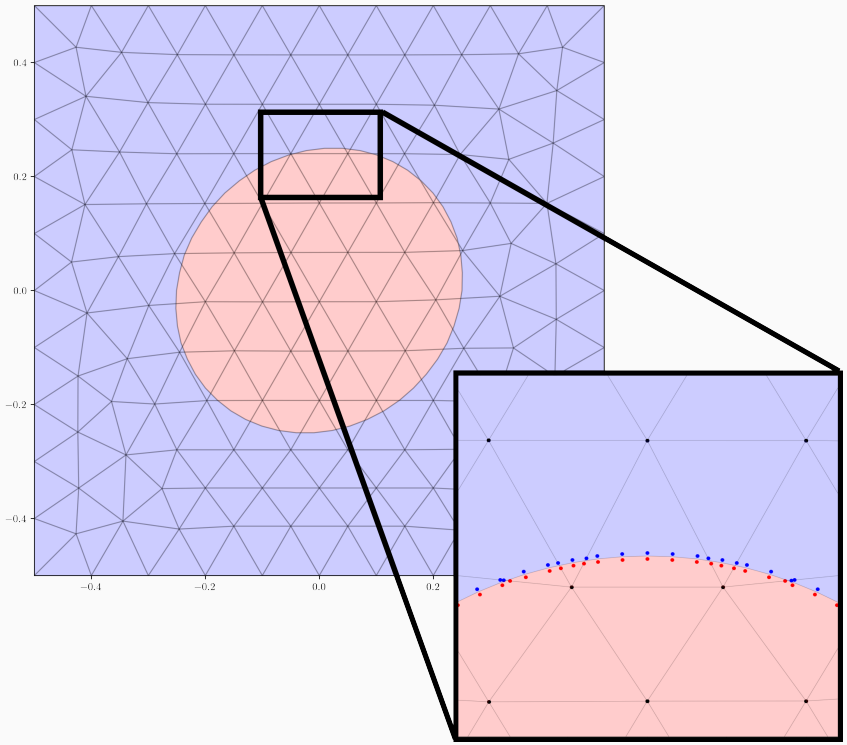
where $[q]_\Gamma := q_{\mathrm{int}} - q_{\mathrm{ext}}$.

Discrete Setting

Polygonal Interface Γ_h;

Polytopal mesh $(\mathcal{T}_h, \mathcal{E}_h, \mathcal{V}_h)$, **fitted**: Γ_h is union of edges in $\mathcal{E}_h$.





Remark: arbitrary precision in discretization of curved interfaces.

Discrete Space

DDR space on $\Omega_\bullet$:

$$\underline{V}^k_{\bullet,h} := \Big\{ \underline{v}_h = \big((v_T)_{T\in\mathcal{T}_h^\bullet}, (v_E)_{E\in\mathcal{E}_h^\bullet}, (v_V)_{V\in\mathcal{V}_h^\bullet}\big) \ : \\ v_T \in \mathcal{P}^{k-1}(T) \text{ for all } T \in \mathcal{T}_h^\bullet, \\ v_E \in \mathcal{P}^{k-1}(E) \text{ for all } E \in \mathcal{E}_h^\bullet, \\ v_V \in \mathbb{R} \text{ for all } V \in \mathcal{V}_h^\bullet \Big\}.$$

Global DDR Space:

$$\underline{V}^k_h := \underline{V}^k_{\mathrm{int},h} \times \underline{V}^k_{\mathrm{ext},h}.$$

Remarks:

- DOFs on shared geometric entities (i.e. on Γ_h) are doubled,
- With $k = 0$, DOFs are only set at vertices (link with FEM at lowest order).

Local Reconstructions

Following [Di Pietro and Droniou, 2023]:
Edge potential $v_{TE} \in \mathcal{P}^{k+1}(E)$ s.t.

$$v_{TE}(x_V) = v_V \text{ for any endpoint } V \text{ of } E \text{ and } \pi_E^{k-1} v_{TE} = v_E;$$

Discrete gradient $G_T^k : \underline{V}_T^k \to \mathcal{P}^k(T)^2$:

$$\int_T G_T^k \underline{v}_T \cdot \tau = -\int_T v_T (\nabla \cdot \tau) + \sum_{E \in \mathcal{E}_T} \omega_{TE} \int_E v_{TE} (\tau \cdot n_E) \quad \forall \tau \in \mathcal{P}^k(T)^2, \tag{2}$$

Potential $p_T^{k+1} : \underline{V}_T^k \to \mathcal{P}^{k+1}(T)$:

$$\int_T p_T^{k+1} \underline{v}_T (\nabla \cdot \tau) = -\int_T G_T^k \underline{v}_T \cdot \tau + \sum_{E \in \mathcal{E}_T} \omega_{TE} \int_E v_{TE} (\tau \cdot n_E)$$
$$\forall \tau \in \mathcal{R}^{c,k+2}(T). \tag{3}$$

Interface Trace Operators

Edge Jump:

$$[\underline{v}_h]_E := v_{T_{\mathrm{int}}E} - v_{T_{\mathrm{ext}}E}; \tag{4}$$

Average:

$$\lambda_{\mathrm{int}} := \frac{\sigma_{\mathrm{ext}}}{\sigma_{\mathrm{int}} + \sigma_{\mathrm{ext}}}, \qquad \lambda_{\mathrm{ext}} := \frac{\sigma_{\mathrm{int}}}{\sigma_{\mathrm{int}} + \sigma_{\mathrm{ext}}}, \tag{5}$$

with:

$$\{\underline{v}_h\}_{\overline{\lambda},E} := \lambda_{\mathrm{ext}} v_{T_{\mathrm{int}}E} + \lambda_{\mathrm{int}} v_{T_{\mathrm{ext}}E}; \tag{6}$$

Skew Average:

$$\{v\}_{\lambda,E} := \lambda_{\mathrm{int}} \gamma_{T_{\mathrm{int}}E} v + \lambda_{\mathrm{ext}} \gamma_{T_{\mathrm{ext}}E} v. \tag{7}$$

Remark: weights in average operators are pivotal to prove robustness w.r.t. σ constrast.

Integration by parts

$$\left(\sum_{\bullet\in\{\mathrm{int},\mathrm{ext}\}}\int_{\Omega_\bullet}\sigma_\bullet\nabla u_\bullet\cdot\nabla v_\bullet\right)-\int_\Gamma\{\sigma_\bullet\nabla u\}_\lambda\cdot n_\Gamma[v]_\Gamma=$$
$$\left(\sum_{\bullet\in\{\mathrm{int},\mathrm{ext}\}}\int_{\Omega_\bullet}fv_\bullet\right)+\int_\Gamma\Phi_\Gamma\{\sigma_\bullet\nabla u\}_{\overline{\lambda}}\cdot n_\Gamma,$$

remarking:

$$\int_\Gamma[\sigma\nabla u\cdot n_\Gamma v]_\Gamma=\int_\Gamma\{\sigma\nabla u\cdot n_\Gamma\}_\lambda[v]_\Gamma+\int_\Gamma[\sigma\nabla u\cdot n_\Gamma]_\Gamma\{v\}_{\overline{\lambda}}.$$

Discrete Problem

Find $\underline{u}_h \in \underline{V}_{h,0}^k$ s.t.:

$$a_h(\underline{u}_h, \underline{v}_h) = \ell_h(\underline{v}_h) \qquad \forall \underline{v}_h \in \underline{V}_{h,0}^k, \tag{8}$$

with:

$$\begin{aligned} a_h(\underline{w}_h, \underline{v}_h) \coloneqq & \sum_{T \in \mathcal{T}_h} \left(\int_T \sigma_T G_T^k \underline{w}_T \cdot G_T^k \underline{v}_T + \frac{\sigma_T}{h_T} \sum_{E \in \mathcal{E}_T} \int_E (p_T^{k+1} \underline{w}_T - w_{TE})(p_T^{k+1} \underline{v}_T - v_{TE}) \right) \\ & - \sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E \{\sigma G_h^k \underline{w}_h\}_{\lambda,E} \cdot n_E [\underline{v}_h]_E + \eta \sum_{E \in \mathcal{E}_{\Gamma,h}} \frac{\alpha}{h_E} \int_E [\underline{w}_h]_E [\underline{v}_h]_E, \end{aligned} \tag{9}$$

where

$$\alpha \coloneqq \frac{2\sigma_{\mathrm{int}} \sigma_{\mathrm{ext}}}{\sigma_{\mathrm{int}} + \sigma_{\mathrm{ext}}}, \tag{10}$$

and

$$\ell_h(\underline{v}_h) \coloneqq \sum_{T \in \mathcal{T}_h} \int_T f p_T^{k+1} \underline{v}_T + \sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E \Phi_\Gamma \{\underline{v}_h\}_{\overline{\lambda},E} + \eta \sum_{E \in \mathcal{E}_{\Gamma,h}} \frac{\alpha}{h_E} \int_E J_\Gamma [\underline{v}_h]_E. \tag{11}$$

Norm and Interpolator

Energy Norm:

$$\|\underline{v}_h\|_{\mathrm{en},h}^2 := \sum_{T\in\mathcal{T}_h} \sigma_T \left(\|G_T^k \underline{v}_T\|_{L^2(T)^2}^2 + h_T^{-1} \sum_{E\in\mathcal{E}_T} \|p_T^{k+1}\underline{v}_T - v_{TE}\|_{L^2(E)}^2 \right) + |\underline{v}_h|_{\mathrm{J},h}^2, \quad (12)$$

with jump seminorm:

$$|\underline{v}_h|_{\mathrm{J},h}^2 := \sum_{E\in\mathcal{E}_{\Gamma,h}} \frac{\alpha}{h_E} \|[\underline{v}_h]_E\|_{L^2(E)}^2; \quad (13)$$

Interpolator $\underline{I}_h^k : C^0(\overline{\Omega}_{\mathrm{int}}) \times C^0(\overline{\Omega}_{\mathrm{ext}}) \to \underline{V}_h^k$ s.t.:

$$\underline{I}_{\bullet,h}^k v_\bullet := \left((\pi_T^{k-1} v_\bullet)_{T\in\mathcal{T}_h^\bullet}, (\pi_E^{k-1} v_\bullet)_{E\in\mathcal{E}_h^\bullet}, (v_\bullet(x_V))_{V\in\mathcal{V}_h^\bullet} \right).$$

Stability

Theorem

Let $N_\partial := \max_{T \in \mathcal{T}_h} \operatorname{card}(\mathcal{E}_T \cap \mathcal{E}_{\Gamma,h})$ and C_{tr} be the trace inequality constant. Assume that

$$\eta > \frac{C_{\mathrm{tr}}^2 N_\partial}{4\epsilon} \tag{14}$$

for some real number $0 < \epsilon < 1$. Then, for all $\underline{v}_h \in \underline{V}_{h,0}^k$, it holds

$$C_{\mathrm{stab}} \|\underline{v}_h\|_{\mathrm{en},h}^2 \leq a_h(\underline{v}_h, \underline{v}_h), \tag{15}$$

with $C_{\mathrm{stab}} := \min\left\{1 - \epsilon, \eta - \frac{C_{\mathrm{tr}}^2 N_\partial}{4\epsilon}\right\}$.

A priori Error Estimate

Theorem

Let $u = (u_{\mathrm{int}}, u_{\mathrm{ext}}) \in \left[C^0(\overline{\Omega}_{\mathrm{int}}) \cap H^{r+2}(\mathcal{T}_h^{\mathrm{int}})\right] \times \left[C^0(\overline{\Omega}_{\mathrm{ext}}) \cap H^{r+2}(\mathcal{T}_h^{\mathrm{ext}})\right]$. Under assumption (14), with $\overline{\sigma} := \max\{\sigma_{\mathrm{int}}, \sigma_{\mathrm{ext}}\}$ and for some $r \in \{0, \ldots, k\}$, it holds:

$$\|\underline{u}_h - \underline{I}_h^k u\|_{\mathrm{en},h} \lesssim \overline{\sigma}^{1/2} h^{r+1} |u|_{H^{r+2}(\mathcal{T}_h)},$$

where $|\cdot|_{H^{r+2}(\mathcal{T}_h)}$ is the broken H^{r+2}-seminorm on the mesh $\mathcal{T}_h$ and the hidden constant depends only on the domain, the stability constant C_{stab}, the polynomial degree k, and the mesh regularity parameter (but is independent of both the meshsize and σ).

Sketch of Proof

$$
\begin{aligned}
a_h(\underline{v}_h, \underline{v}_h) \coloneqq & \sum_{T \in \mathcal{T}_h} \left(\int_T \sigma_T G_T^k \underline{v}_T \cdot G_T^k \underline{v}_T + \frac{\sigma_T}{h_T} \sum_{E \in \mathcal{E}_T} \int_E (p_T^{k+1} \underline{v}_T - v_{TE})(p_T^{k+1} \underline{v}_T - v_{TE}) \right) \\
& - \sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E \{\sigma G_h^k \underline{v}_h\}_{\lambda,E} \cdot n_E [\underline{v}_h]_E + \eta \sum_{E \in \mathcal{E}_{\Gamma,h}} \frac{\alpha}{h_E} \int_E [\underline{v}_h]_E [\underline{v}_h]_E \\
= & \sum_{T \in \mathcal{T}_h} \sigma_T (\| G_T^k \underline{v}_T \|^2_{L^2(T)^2} + h_T^{-1} \sum_{E \in \mathcal{E}_T} \| p_T^{k+1} \underline{v}_T - v_{TE} \|^2_{L^2(E)}) \\
& - \sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E \{\sigma G_h^k \underline{v}_h\}_{\lambda,E} \cdot n_E [\underline{v}_h]_E + \eta |\underline{v}_h|^2_{\mathrm{J},h},
\end{aligned}
\tag{16}
$$

Recall definition of norm:

$$
\| \underline{v}_h \|^2_{\mathrm{en},h} \coloneqq \sum_{T \in \mathcal{T}_h} \sigma_T \left(\| G_T^k \underline{v}_T \|^2_{L^2(T)^2} + h_T^{-1} \sum_{E \in \mathcal{E}_T} \| p_T^{k+1} \underline{v}_T - v_{TE} \|^2_{L^2(E)} \right) + |\underline{v}_h|^2_{\mathrm{J},h}, \tag{17}
$$

Sketch of Proof

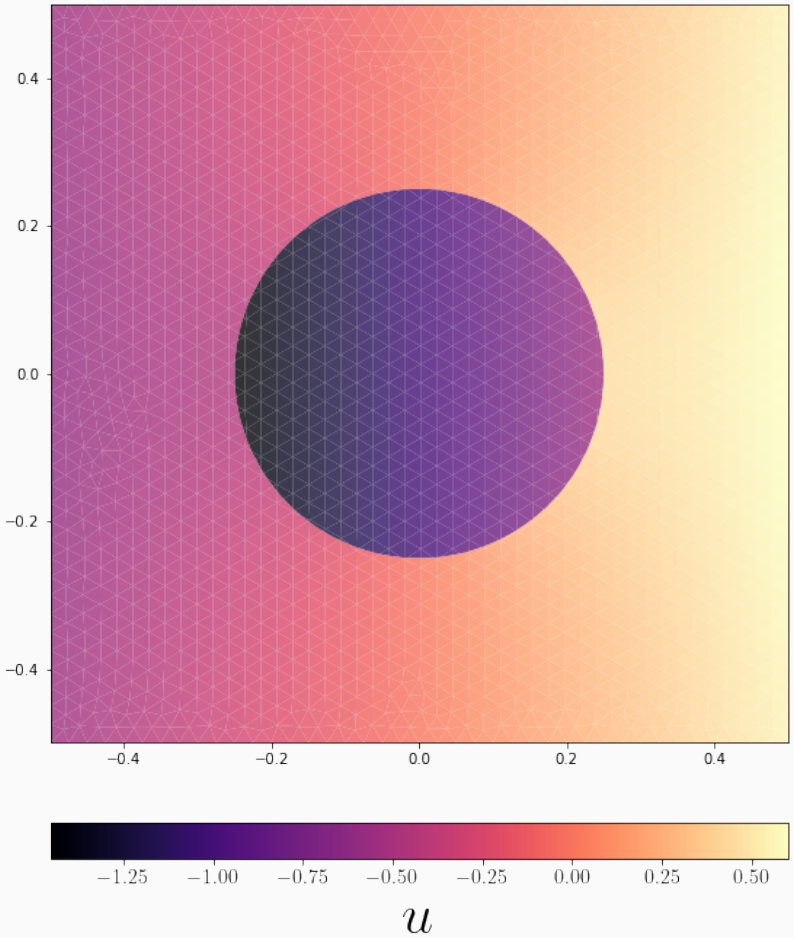
$$
\begin{aligned}
&\int_E \{\sigma G_h^k \underline{w}_h\}_{\lambda,E} \cdot n_E [\underline{v}_h]_E \\
&\leq \|\{\sigma G_h^k \underline{w}_h\}_{\lambda,E}\|_{L^2(E)^2} \|n_E\|_{L^\infty(E)^2} \|[\underline{v}_h]_E\|_{L^2(E)} \\
&\leq \alpha^{-1/2} \left(\lambda_{\mathrm{int}} \sigma_{\mathrm{int}}^{1/2} \|\sigma_{\mathrm{int}}^{1/2} G_{T_{\mathrm{int}}}^k \underline{w}_{T_{\mathrm{int}}}\|_{L^2(E)^2} + \lambda_{\mathrm{ext}} \sigma_{\mathrm{ext}}^{1/2} \|\sigma_{\mathrm{ext}}^{1/2} G_{T_{\mathrm{ext}}}^k \underline{w}_{T_{\mathrm{ext}}}\|_{L^2(E)^2}\right) \ \alpha^{1/2} \|[\underline{v}_h]_E\|_{L^2} \\
&\leq \frac{1}{\sqrt{2}} \left(\|\sigma_{\mathrm{int}}^{1/2} G_{T_{\mathrm{int}}}^k \underline{w}_{T_{\mathrm{int}}}\|_{L^2(E)^2} + \|\sigma_{\mathrm{ext}}^{1/2} G_{T_{\mathrm{ext}}}^k \underline{w}_{T_{\mathrm{ext}}}\|_{L^2(E)^2}\right) \ \alpha^{1/2} \|[\underline{v}_h]_E\|_{L^2(E)} \\
&\leq \frac{C_{\mathrm{tr}}}{\sqrt{2}} \left(\|\sigma_{\mathrm{int}}^{1/2} G_{T_{\mathrm{int}}}^k \underline{w}_{T_{\mathrm{int}}}\|_{L^2(T_{\mathrm{int}})^2} + \|\sigma_{\mathrm{ext}}^{1/2} G_{T_{\mathrm{ext}}}^k \underline{w}_{T_{\mathrm{ext}}}\|_{L^2(T_{\mathrm{ext}})^2}\right) \ \left(\frac{\alpha}{h_E}\right)^{1/2} \|[\underline{v}_h]_E\|_{L^2(E)},
\end{aligned}
$$

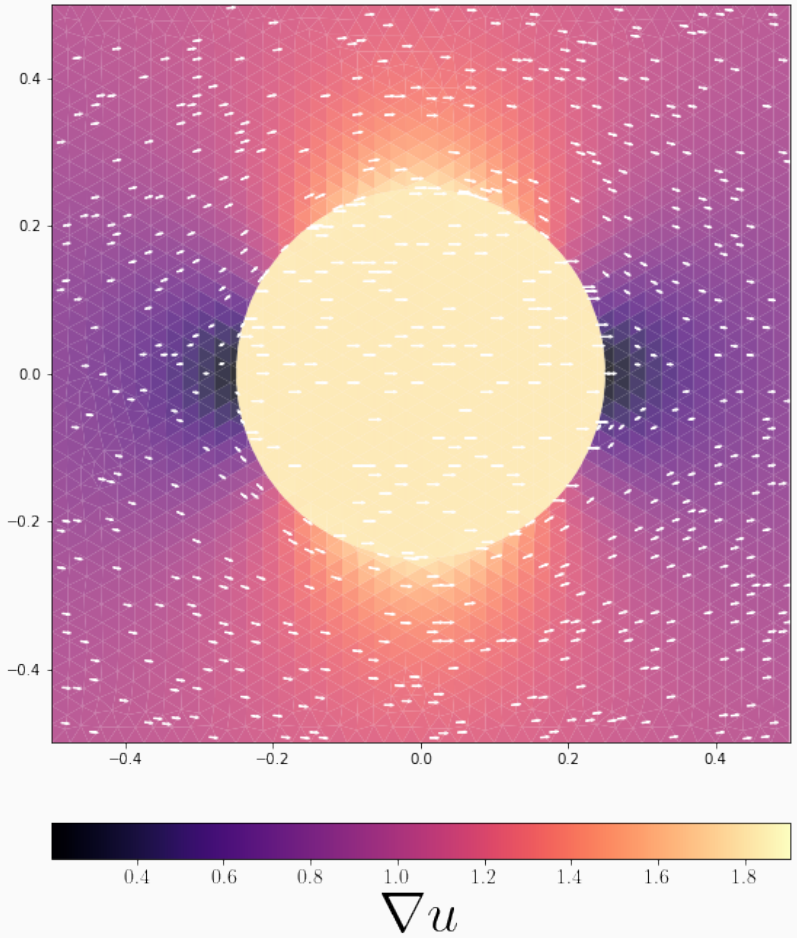
exploiting:

$$
\frac{\lambda_\bullet^2 \sigma_\bullet}{\alpha} < \frac{1}{2} \qquad \bullet \in \{\mathrm{int}, \mathrm{ext}\}. \tag{18}
$$

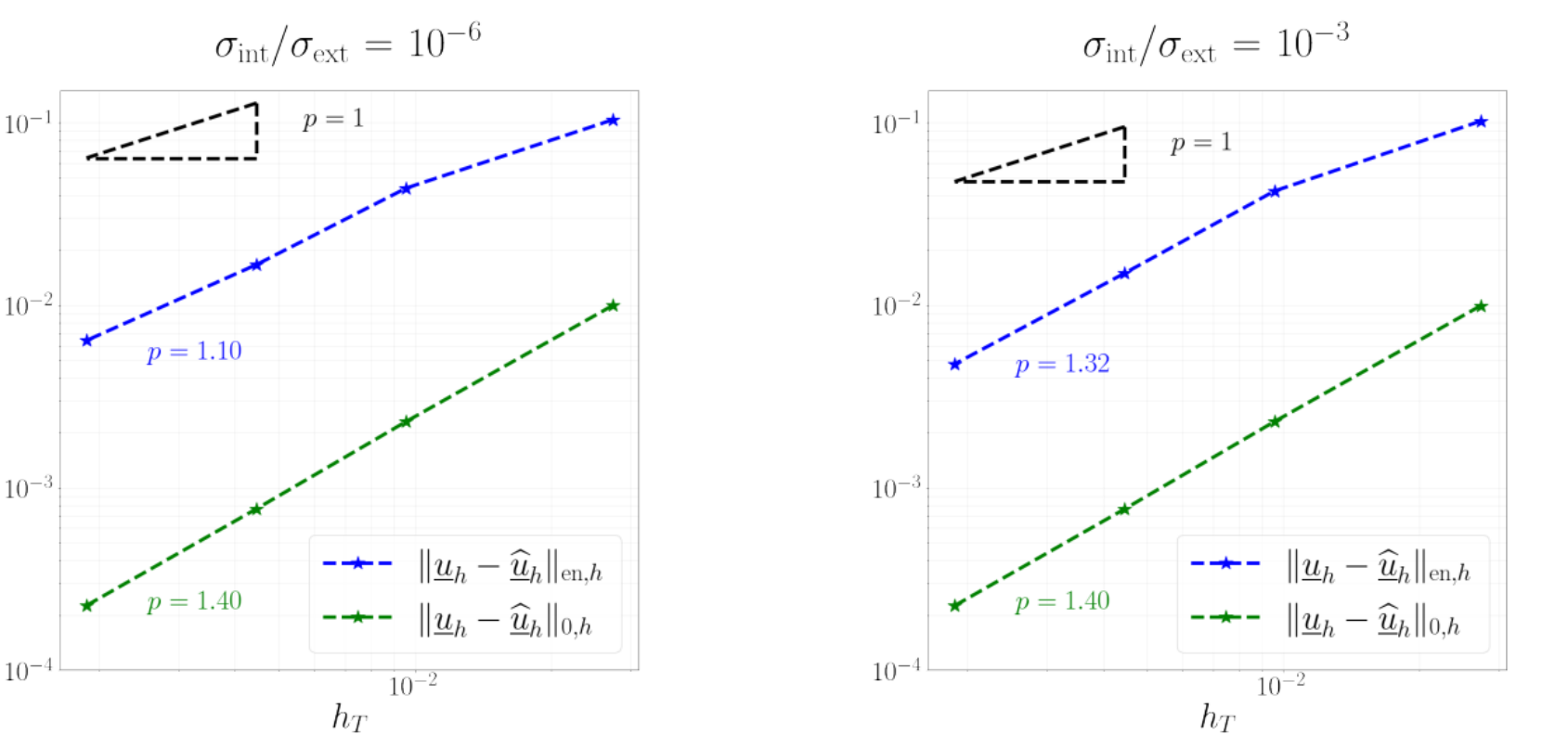
Numeric Test

$$u_{\text{int}} = \frac{2\sigma_{\text{ext}}}{\sigma_{\text{ext}} + \sigma_{\text{ext}}} x, \qquad u_{\text{ext}} = 1 + \left[1 + \left(\frac{\sigma_{\text{ext}} - \sigma_{\text{int}}}{\sigma_{\text{ext}} + \sigma_{\text{int}}}\right) \frac{R^2}{x^2 + y^2}\right] x$$



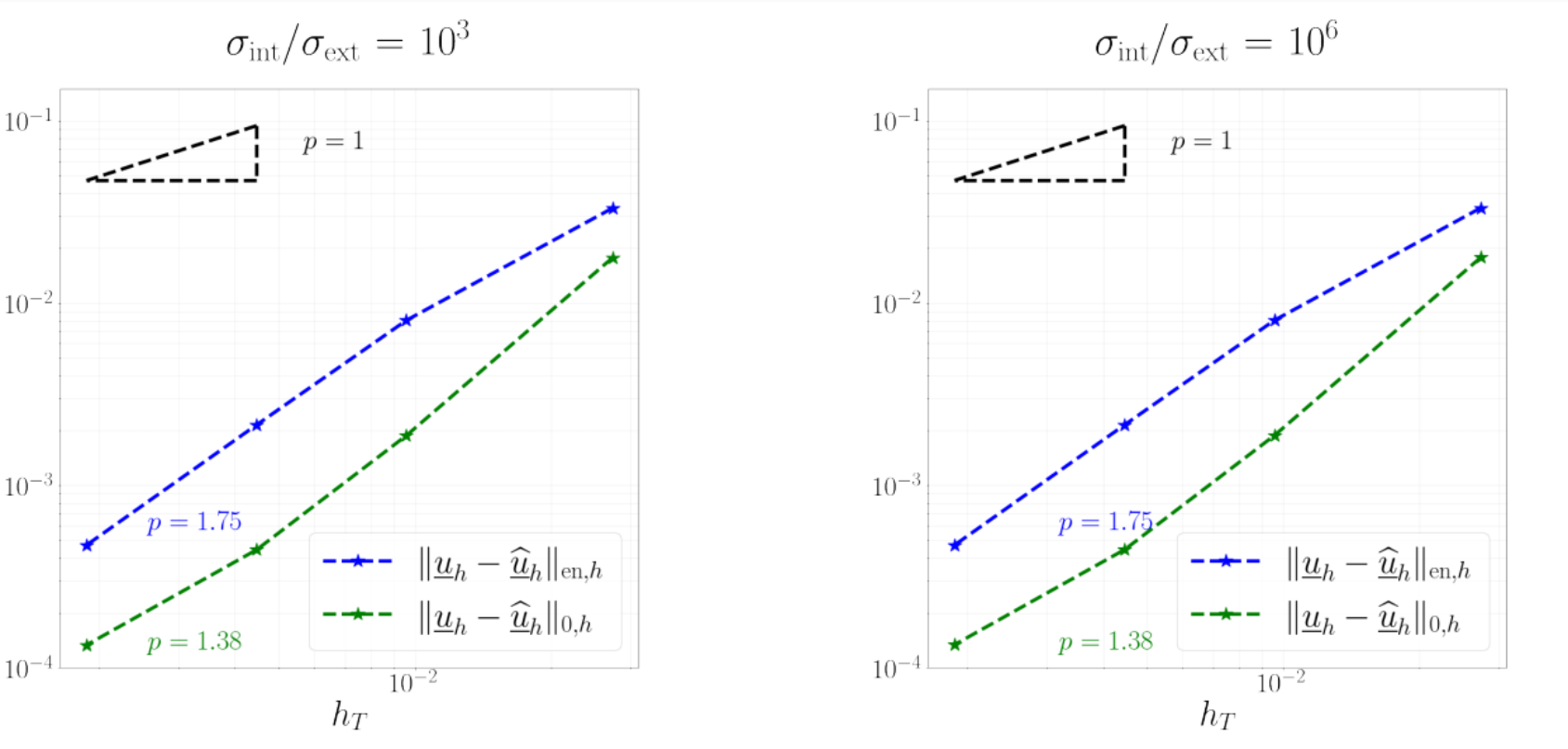


Convergence Trends, $k = 0$



$$\|\underline{v}_h\|_{0,h} := \left(\sum_{T \in \mathcal{T}_h} \|v_T\|^2_{L^2(T)} + h_T \sum_{E \in \mathcal{E}_T} \|v_E\|^2_{L^2(E)} \right)^{\frac{1}{2}} \qquad \forall \underline{v}_h \in \underline{V}_h^k.$$

Convergence Trends, $k = 0$



Time-dependent J_Γ

Given $f : (0, t_\mathrm{f}] \to \mathbb{R}$, J_Γ^0: find $u = (u_\mathrm{int}, u_\mathrm{ext})$ and $J_\Gamma : \Gamma \times [0, t_\mathrm{f}] \to \mathbb{R}$ such that

$$
\begin{aligned}
-\nabla \cdot (\sigma_\bullet \nabla u_\bullet) &= f && \text{in } \Omega_\bullet \times (0, t_\mathrm{f}],\ \bullet \in \{\mathrm{int}, \mathrm{ext}\}, && \text{(19a)}\\
[u]_\Gamma &= J_\Gamma && \text{on } \Gamma \times (0, t_\mathrm{f}], && \text{(19b)}\\
[\sigma \nabla u]_\Gamma \cdot n_\Gamma &= 0 && \text{on } \Gamma \times (0, t_\mathrm{f}], && \text{(19c)}\\
u_\mathrm{ext} &= 0 && \text{on } \partial\Omega \times (0, t_\mathrm{f}] && \text{(19d)}\\
C \partial_t J_\Gamma &= \sigma_\bullet \nabla u_\bullet \cdot n_\Gamma && \text{on } \Gamma \times (0, t_\mathrm{f}],\ \bullet \in \{\mathrm{int}, \mathrm{ext}\} && \text{(19e)}\\
J_\Gamma(\cdot, 0) &= J_\Gamma^0 && \text{on } \Gamma, && \text{(19f)}
\end{aligned}
$$

with $C > 0$.

Discretization of Jump Evolution

Time discretization of (19e):

$$\frac{C}{\tau}\left(J_\Gamma^{n+1} - J_\Gamma^n\right) = \sigma_{\text{int}} \nabla u_{\text{int}}^n \cdot n_\Gamma = \sigma_{\text{ext}} \nabla u_{\text{ext}}^n \cdot n_\Gamma.$$

Space of Jumps:

$$\underline{Z}_{\Gamma,h} := \left\{ \underline{p}_{\Gamma,h} = (p_V)_{V \in \mathcal{V}_{\Gamma,h}} \ : \ p_V \in \mathbb{R} \quad \forall V \in \mathcal{V}_{\Gamma,h} \right\}.$$

Advancement of Jump: find $\underline{J}_{\Gamma,h}^{n+1} \in \underline{Z}_{\Gamma,h}$ such that

$$\frac{C}{\tau}\left(\sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E J_E^{n+1} Q_E^n - \int_E J_E^n Q_E^n \right) = \sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E \{\sigma G_T \underline{u}_T^n\}_{\lambda,E} \cdot n_\Gamma Q_E \qquad \forall \underline{Q}_{\Gamma,h}^n \in \underline{Z}_{\Gamma,h}. \tag{20}$$

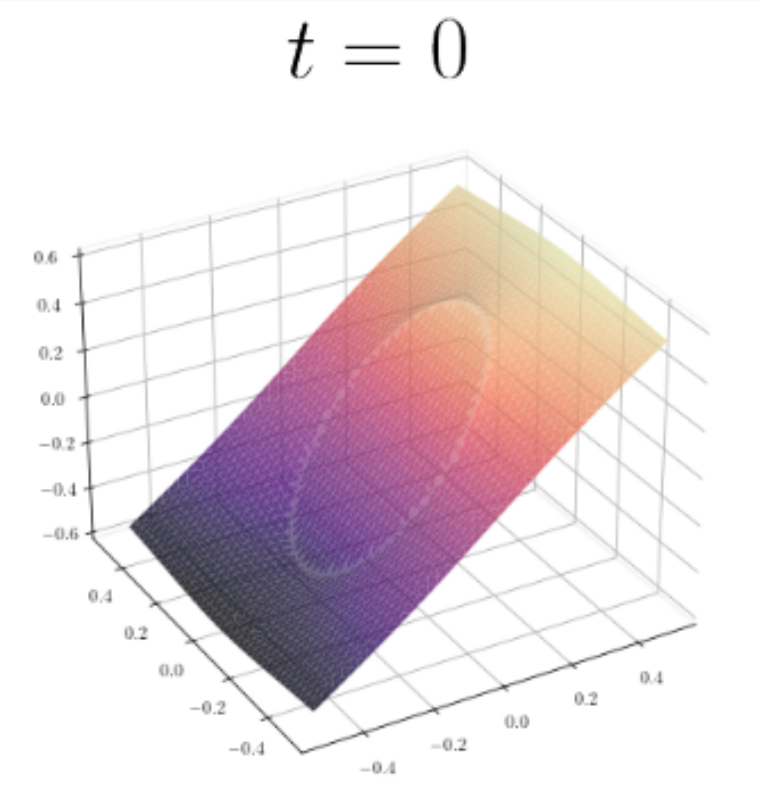
Time Advancement

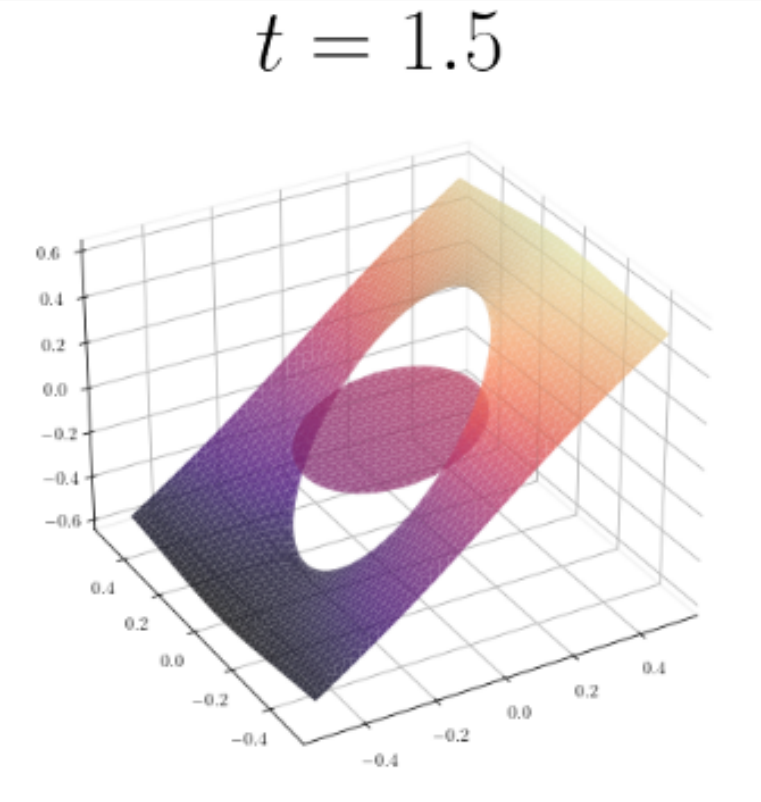
Given $\underline{J}^0_{\Gamma,h}$, for $n = 0, \ldots, N-1$, set, in this order,

$$\underline{u}^n_h \leftarrow \underline{J}^n_\Gamma,$$
$$\underline{J}^{n+1}_{\Gamma,n} \leftarrow \underline{u}^n_h, \underline{J}^n_{\Gamma,h}.$$

Numeric Test

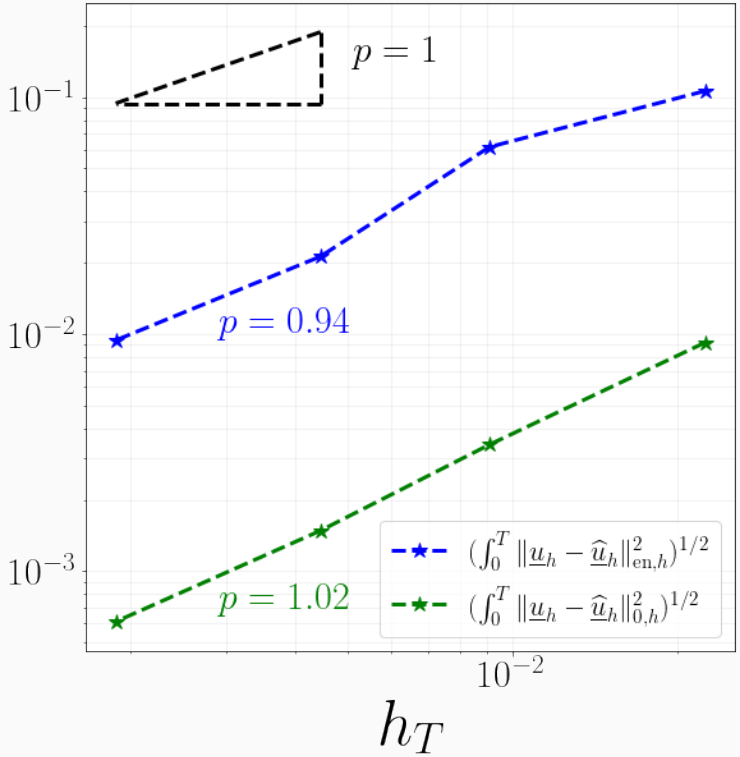
$$\widehat{u}_\bullet(x,t) = \exp\left(-\frac{t}{t_c}\right)\left(u_\bullet^0(x) - u_\bullet^\infty(x)\right) + u_\bullet^\infty(x) \tag{21}$$

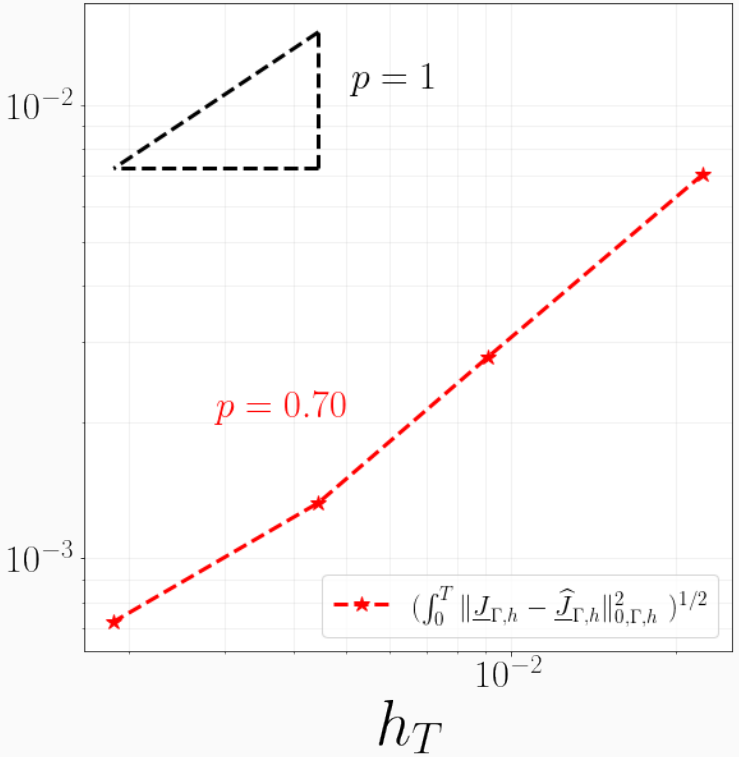




Convergence Trend, $k = 0$

$$\|J_{\Gamma,h}\|^2_{0,\Gamma,h} := \sum_{E \in \mathcal{E}_{\Gamma,h}} \|J_E\|^2_E.$$





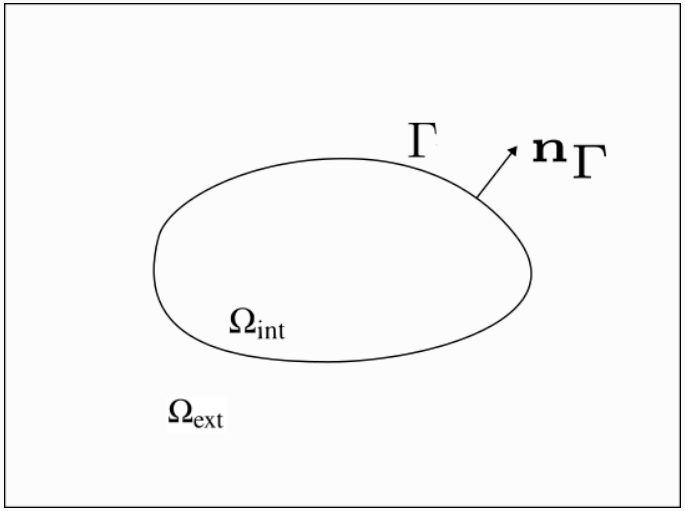
Moving Membrane

Dynamical Process of the Membrane:

$$\Gamma(t) : \boldsymbol{x}(s,t).$$

Membrane Advection:

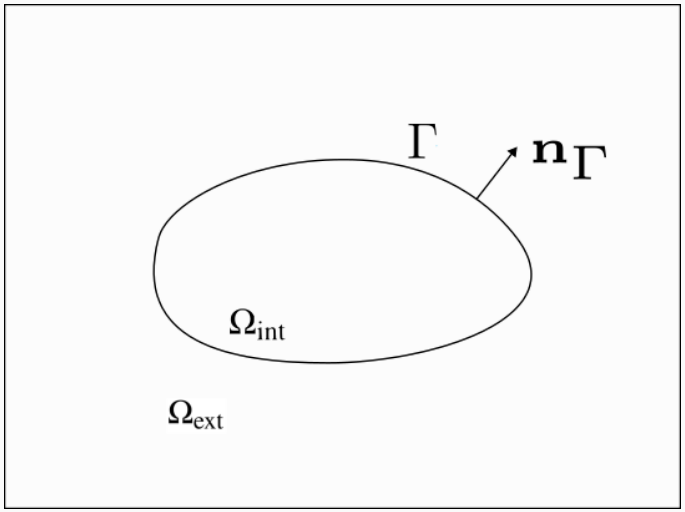
$$\dot{\boldsymbol{x}}(s) = \boldsymbol{v}(\boldsymbol{x}(s))$$



Stationary Stokes with Interface Conditions

Find $\boldsymbol{v} : \Omega \to \mathbb{R}^d$ and $p : \Omega_{\text{int}} \cup \Omega_{\text{ext}} \to \mathbb{R}$ s.t.:

$$\begin{cases} \nabla \cdot \sigma = 0 & \text{in } \Omega_\bullet \\ \nabla \cdot \boldsymbol{v} = 0 & \text{in } \Omega_\bullet \\ [v]_E = 0 & \text{on } \Gamma \\ [\sigma]_E \cdot n_\Gamma = \boldsymbol{t}_\Gamma & \text{on } \Gamma \\ \int_\Omega p = 0 \\ \text{BCs} \end{cases}$$



with:

$$\sigma = 2\mu \nabla_s \boldsymbol{v} - p \boldsymbol{I}_d,$$

$$\boldsymbol{t}_\Gamma = \text{tension(strain, curvature, electric field)}$$

References

[1] D. A. Di Pietro and J. Droniou. **"An arbitrary-order discrete de Rham complex on polyhedral meshes: Exactness, Poincaré inequalities, and consistency".** In: *Found. Comput. Math.* 23 (2023), pp. 85–164. DOI: 10.1007/s10208-021-09542-8.